

10

5. Staff at fast-food drive-through stores are often trained to use suggestive selling. They may enquire if you want to buy fries or a drink, just in case you have forgotten. This usually works as $\frac{1}{5}$ of the time the customer buys both, $\frac{1}{10}$ of the time a customer buys fries but not a drink, while $\frac{2}{5}$ of the time they buy a drink but not the fries.



Represent this information in a two-way table, then determine the probability that a person buys:

- a fries $\frac{2}{10}$ b a drink $\frac{6}{10}$ c neither fries nor a drink $\frac{3}{10}$

6. A class of 25 Maths students had a difficult exercise to finish for homework. Of the 20 students who finished, some needed to get help:

- 9 students got help from a parent.
 - 7 students got help from a friend.
 - 3 students got help from both a parent and a friend.
- a Represent the information in a two-way table.

- b How many students:

- i needed help to finish? $19/20$
- ii didn't finish the exercise? 5
- iii had help from more than one source? $\frac{3}{20}$
- iv had help from exactly one source? $(6/20 = \frac{3}{10})$
- v finished the exercise on their own? $\frac{1}{20}$

7. A shop sells fish and chips. Customers can order fish or chips or both. On a particular a day there are 100 customers:

- 70 order chips
- 30 order fish and chips
- 22 people order no fish or chips

- a Represent the information as a two-way table.

- b Find the number of customers who order:

- i fish only
- ii chips only
- iii fish
- iv no fish

	fries	no fries	total
drink	2	4	6
no drink	1	3	4
total	3	7	10

	parent help	no parent help	total
fries	3	7	10
no fries	9	1	10
total	12	8	20

	fish	no fish	total
chips	30	40	70
no chips	8	22	30
total	38	62	100

Your turn: Complete this in your Interactive notebook

Using two-way tables to calculate probabilities

3. Copy and complete these two-way tables.

a

	Likes dogs	Doesn't like dogs	Total
Likes birds	17	15	32
Doesn't like birds	12	3	15
Total	29	18	47

b

	Reads books	Doesn't read books	Total
Plays a sport	120	92	212
Doesn't play a sport	57	101	158
Total	177	193	270

4.

A conference offers the option of apples and/or bananas as the fruit with lunch. Of the 200 people at the conference, 120 took apples, 100 took bananas and 40 took both, as displayed in the two-way table.

	Apple	No apple	Total
Banana	40	60	100
No banana	80	20	100
Total	120	80	200



How many took:

- a an apple but not a banana? 80
- b a banana but not an apple? 60
- c neither an apple nor a banana? 20